

SOC Home Lab

Part 3 – Deployment of Wazuh Virtual Machine and agent on Windows Endpoint

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Technologies Used

- Oracle VirtualBox
- Windows 11 Microsoft Sysmon
- Wazuh

OBJECTIVE:

The objective of this lab is to install and deploy a Wazuh SIEM server and enroll a Windows 11 endpoint to centralize endpoint telemetry for monitoring, alerting, and future security analysis.

Lab Setup

Wazuh virtual machine image was downloaded, installed, and deployed on Oracle VirtualBox. During Wazuh start-up, administrator credentials were inputted and verified, while logged in as administrator Wazuh dashboard was retrieved. Powered on Windows 11 virtual machine, deployed Wazuh on Windows endpoint via command.

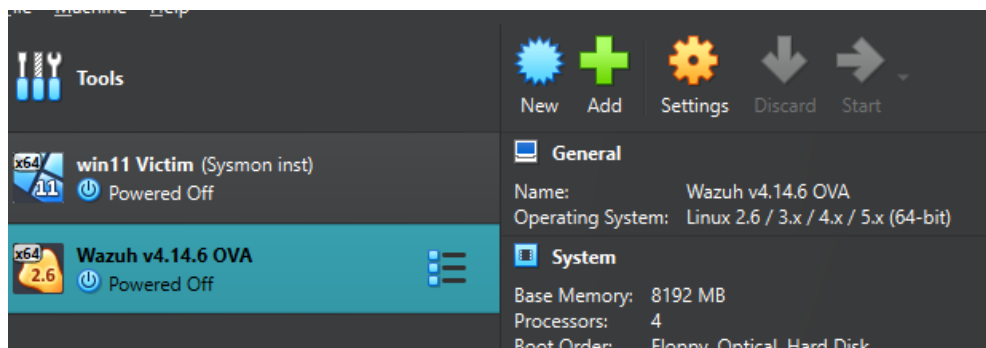


Figure 1. Successful installation of Wazuh on VirtualBox



Figure 2. Wazuh dashboard authentication login

Troubleshooting

Problem

The Wazuh Dashboard displayed:

API connections could be down or inaccessible.

The dashboard reported the Wazuh API status as Offline.

Investigation

Verified:

- Network connectivity
- VirtualBox adapter configuration
- Wazuh Manager service status

Initially suspected a networking issue because the Windows endpoint could not deploy the Wazuh agent.

Resolution

Restarted the Wazuh Manager

“sudo systemctl restart Wazuh- manager”

After restarting the Wazuh Manager service, the Dashboard successfully re-established communication with the API and the manager status changed from Offline to Online.

Verified:

“sudo systemctl status Wazuh-manager”

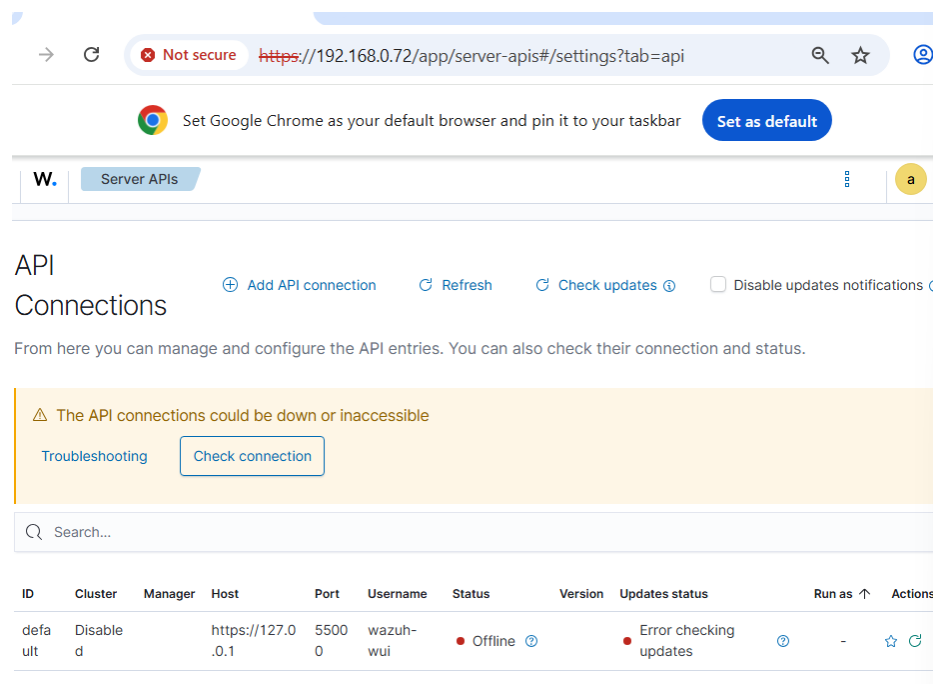


Figure 3. Wazuh Dashboard displays API connection Status as down and inaccessible, indicating that Dashboard is unable to talk to Wazuh API.

```

lwazuh-user@wazuh-server ~$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-07-13 03:11:27 UTC; 33s ago
     Process: 2660 ExecStart=/usr/bin/envu /var/ossec/bin/wazuh-control start (code=exited, status=0)
    Tasks: 233 (limit: 9477)
   Memory: 752.2M
      CPU: 51.508s
   CGroup: /system.slice/wazuh-manager.service
           └─2428 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             └─2434 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
               └─2435 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                 └─2442 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─2447 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─2758 /var/ossec/bin/wazuh-authd
                       └─2775 /var/ossec/bin/wazuh-db
                         └─2815 /var/ossec/bin/wazuh-execd
                           └─2834 /var/ossec/bin/wazuh-analysd
                             └─2848 /var/ossec/bin/wazuh-syscheckd
                               └─2870 /var/ossec/bin/wazuh-remoted
                                 └─2884 /var/ossec/bin/wazuh-logcollector
                                   └─2898 /var/ossec/bin/wazuh-monitord
                                     └─2919 /var/ossec/bin/wazuh-modulesd
                                       └─4581 sh -c "/bin/ps -p 2471 > /dev/null 2>&1"
                                         └─4583 sh -c "/bin/ps -p 2471 > /dev/null 2>&1"

Jul 13 03:11:24 wazuh-server envu[2660]: wazuh-logcollector: Process 2817 not used by Wazuh, removing...
Jul 13 03:11:24 wazuh-server envu[2660]: Started wazuh-logcollector...
Jul 13 03:11:24 wazuh-server envu[2660]: wazuh-monitord: Process 2843 not used by Wazuh, removing...
Jul 13 03:11:24 wazuh-server envu[2660]: Started wazuh-monitord...
Jul 13 03:11:24 wazuh-server envu[2660]: wazuh-modulesd: Process 2872 not used by Wazuh, removing...
Jul 13 03:11:24 wazuh-server envu[2917]: 2026/07/13 03:11:24 wazuh-modulesd:router: INFO: Loaded router...
Jul 13 03:11:24 wazuh-server envu[2917]: 2026/07/13 03:11:24 wazuh-modulesd:content_manager: INFO: Loaded content manager...
Jul 13 03:11:24 wazuh-server envu[2917]: 2026/07/13 03:11:24 wazuh-modulesd:inventory-harvester: INFO: Loaded inventory harvester...
Jul 13 03:11:25 wazuh-server envu[2660]: Started wazuh-modulesd...
Jul 13 03:11:27 wazuh-server envu[2660]: Completed.
lines 1-35/35 (END)

```

Figure 4. Displays Wazuh managers online and enable after restart, initially Wazuh server displayed manager API was offline.

Wazuh

Not secure <https://192.168.0.72/app/server-apis#/settings?tab=api>

Set Google Chrome as your default browser and pin it to your taskbar [Set as default](#)

W. Server APIs

API Connections

From here you can manage and configure the API entries. You can also check their connection and status.

Search...

ID	Cluster	Manager	Host	Port	Username	Status	Version	Updates status	Run as	Actions
default	Disabled	wazuh-server	https://127.0.0.1	55000	wazuh-wui	Online		Error checking updates	✓	★ ↻

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Figure 5. Successfully displays Wazuh manager and API back to online and available status after troubleshooting and restarting on Wazuh server. Now will be able to access API and deploy Wazuh agent on Windows 11 VM.

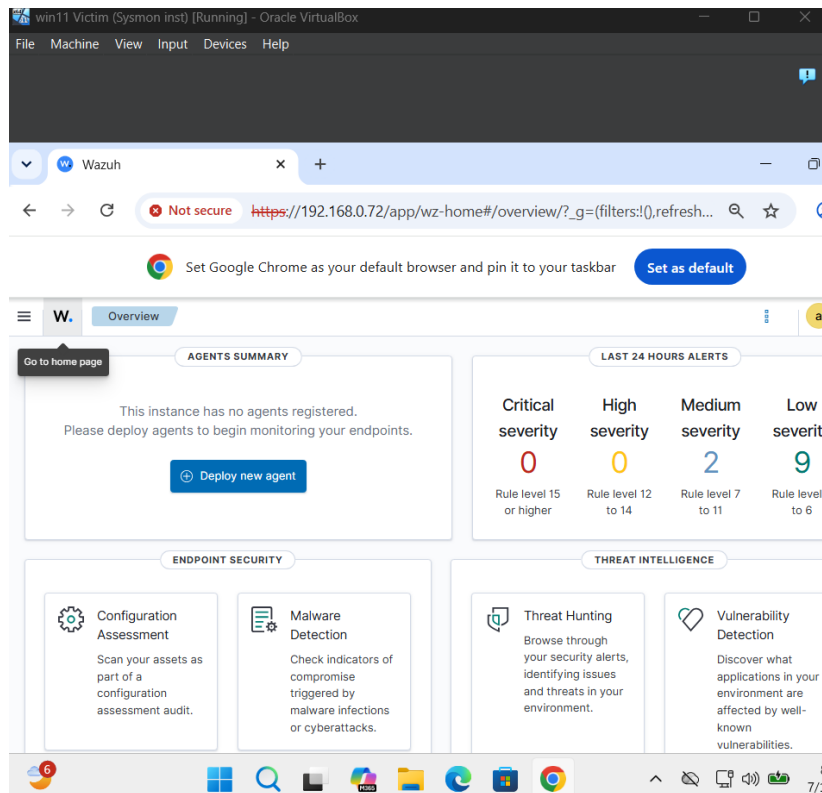


Figure 6. API connection has been achieved, dashboard currently displays 0 agents being deployed to the Windows 11 VM.

Deployment Verification

The Windows 11 endpoint was successfully enrolled into the Wazuh manager. After installing and starting the Wazuh Agent service, the endpoint appeared in the Wazuh Dashboard with an active status, confirming successful communication between the endpoint and the manager. This completed deployment phase and established the foundation for the centralized endpoint monitoring and security event collection.

This verified successful communication between the Wazuh Manager and the Windows endpoint, enabling centralized log collection for future detection and analysis.

```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.6-1.msi -OutFile $env:tmp\wazuh-agent; msexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='192.168.0.72' WAZUH_AGENT_NAME='Win11'
PS C:\WINDOWS\system32> NET START Wazuh
The Wazuh service is starting.
The Wazuh service was started successfully.
PS C:\WINDOWS\system32>
```

Figure 7. PowerShell executed as administrator in order to install Wazuh agent on Windows 11 endpoint.

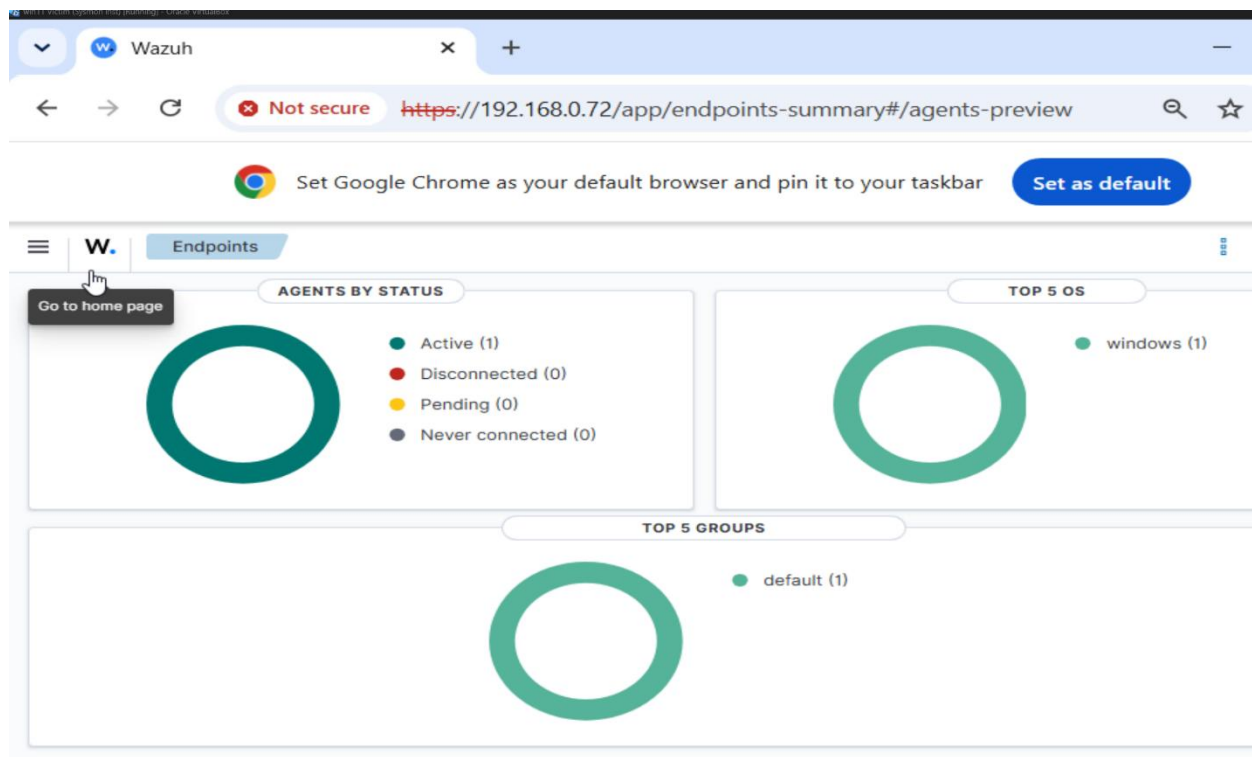


Figure 8. Wazuh manager dashboard displays successful installation of (1) active agent

and connection with (1). Windows 11 OS endpoint.

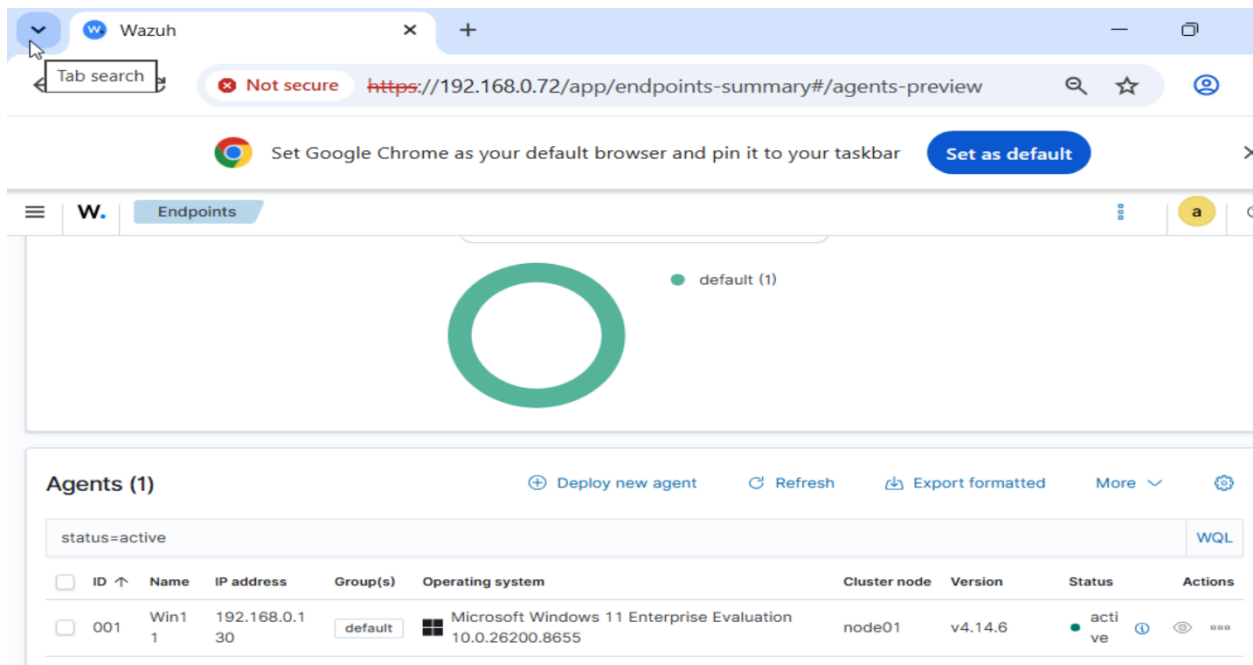


Figure 9. Installed Wazuh Agent correctly displays information on endpoint such as the IP address and OS of Windows 11 VM.

In part 4 I will simulate reconnaissance activity using Kali Linux and analyze the resulting security events with Wazuh.

Skills Demonstrated

- Oracle VirtualBox
- PowerShell
- Wazuh server configuration on Oracle VirtualBox
- Wazuh agent deployment and configuration

Key Takeaways

- Successfully deployed and configured Wazuh server on VirtualBox, providing SIEM like-capabilities centralizing logs and events for future agent deployments.
- Successfully deployed assigned and deployed Wazuh agent on Windows 11 Virtual Machine, providing additional layer of event documentation on top of Sysmon configuration, improving network monitoring capabilities.